

MIAE 221 · Rapid Review Sheet · Part 2

Chemical Bonding, Potential Energy Wells & Physical Properties

1. Primary Chemical Bonds

Bond Type	Electron Mechanism	Directionality	Typical Materials	Key Properties
Ionic	Electron transfer (Metal → Nonmetal)	Non-directional	NaCl, MgO, Al ₂ O ₃	Hard, high T_m , brittle, electrical insulator (solid)
Covalent	Electron sharing (comparable ΔX)	Highly Directional	Diamond, Si, SiC, polymer chains	Very high hardness, high T_m , directional cleavage
Metallic	Valence electron pool ("sea of e^- ")	Non-directional	Fe, Al, Cu, Ti, brass	High ductility, high thermal/electrical cond., luster

2. Pauling Percent Ionic Character (%IC)

% Ionic Character = $[1 - \exp(-0.25(X_A - X_B)^2)] \times 100\%$

- If $\Delta X = |X_A - X_B| = 0 \implies \%IC = 0\%$ (Pure Covalent, e.g. Diamond).
- If $\Delta X \approx 1.7 \implies \%IC \approx 50\%$ (Half Ionic / Half Covalent).
- If $\Delta X > 2.0 \implies \%IC > 63\%$ (Predominantly Ionic, e.g. $TiO_2 = 63.2\%$, $NaCl = 70.3\%$, $CsCl = 75.3\%$).

3. Secondary (Intermolecular) Bonds (4 – 40 kJ/mol)

- Fluctuating Induced Dipoles (London Dispersion):** Instantaneous electron imbalances in symmetrical atoms (e.g. liquid Ar, solid Xenon). Weakest bond (< 10 kJ/mol).
- Permanent Dipoles:** Asymmetrical charge distribution in polar molecules (e.g. HCl).
- Hydrogen Bonding:** Hydrogen covalently bonded to F, O, or N. Bare proton produces uniquely strong secondary bond (10 – 40 kJ/mol). Governs water boiling point, DNA structure, and nylon chain strength.

4. The Potential Energy Well $\longleftrightarrow$ Macroscopic Properties

Deep & Steep Well (Ceramics, W)	Shallow & Broad Well (Polymers, Pb)
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- High Bonding Energy (E_0 large) - High Melting Temperature (T_m) - High Elastic Modulus (Stiff) - Low Thermal Expansion (α small)	- Low Bonding Energy (E_0 small) - Low Melting Temperature (T_m) - Low Elastic Modulus (Compliant) - High Thermal Expansion (α large)

- Melting Temperature (T_m):** Directly proportional to well depth: $T_m \propto |E_0|$.
- Elastic Modulus (E , Stiffness):** Proportional to curve curvature at equilibrium: $E \propto \frac{d^2E_N}{dr^2}\bigg|_{r_0} \propto \frac{dF}{dr}\bigg|_{r_0}$.
- Thermal Expansion (α_l):** Governed by **well asymmetry (anharmonicity)**:
 - Thermal vibration ($E_1 \rightarrow E_2$) pushes atoms wider on attractive side $\implies$ average separation r_{mean} shifts outward.
 - Deep, symmetric well $\implies$ small outward shift $\implies \alpha_l \propto \frac{1}{|E_0|}$.

5. Master Property Matrix

Material Class	Primary Bond	Well Depth $ E_0 $	Modulus (E)	Melting Point	Thermal Exp. (α_l)
Ceramics	Ionic / Covalent	Very Large	150 – 500 GPa	High (> 2000°C)	Low ($1 - 10 \times 10^{-6}/K$)
Metals	Metallic	Moderate–Large	50 – 400 GPa	Moderate–High	Moderate ($10 - 25 \times 10^{-6}/K$)
Polymers	Secondary (chains)	Small	0.1 – 5 GPa	Low (100 – 300°C)	High ($50 - 200 \times 10^{-6}/K$)